

GLearn Artificial Intelligence Laboratory

Assignment 4: State Space Search - Monkey and Banana Problem

1. Problem Formulation and State Representation:

In the Monkey and Banana problem, an agent (the monkey) is placed in a room with a box/chair and a bunch of bananas hanging from the ceiling. The state is represented as a 4-tuple: (Monkey Position, Chair Position, Monkey on Chair, Monkey Holding Banana).

Initial State:

```
initial_state = ('Far-Chair', 'Chair-Not-Under-Banana', 'Off-Chair', 'Empty')
```

Goal State:

```
goal_state = ('Near-Chair', 'Chair-Under-Banana', 'On-Chair', 'Holding')
```

2. Algorithm Used to Explore States:

The algorithm used to explore the states is Breadth-First Search (BFS).

The implementation uses a FIFO queue to systematically explore nodes level-by-level, ensuring the shortest action sequence.

3. Valid Operators and Transitions:

- Walk(x, y): Move the monkey from location x to location y.
- Push(x, y): Push the chair from location x to location y.
- ClimbOn(): Climb onto the chair.
- Grasp(): Grasp the bananas when on the chair directly under the bananas.

4. Implementation Details:

The Python program executes `bfs(initial_state, goal_state)` using a visited set to avoid cycles and explores 14 state transitions.